Imbalance-Aware Feature Selection and Machine Learning for U.S. Corporate Bankruptcy Prediction

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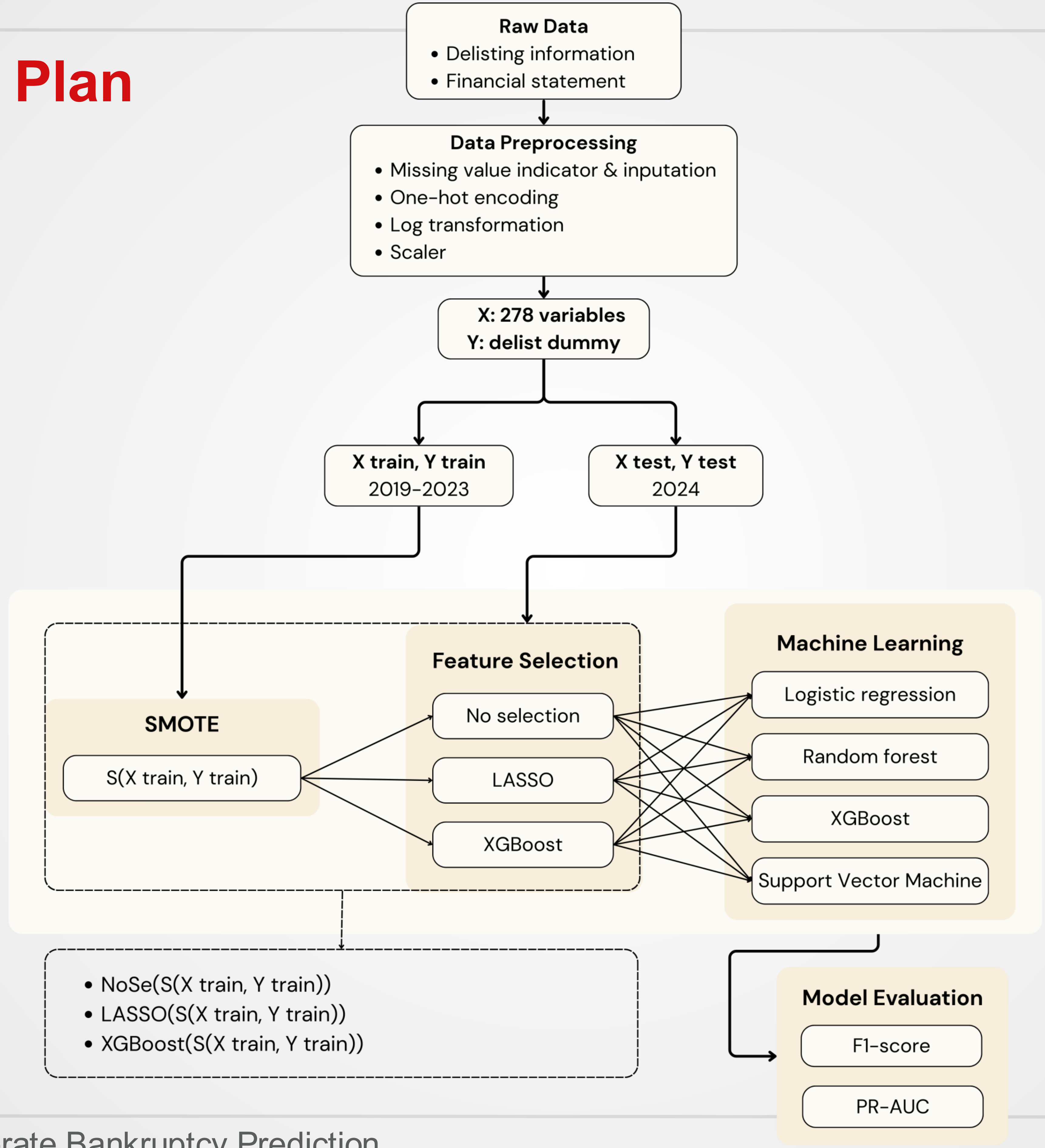
Outline

1. Motivation
2. Study Plan
3. Data (EDA)
4. Model Result
   - LR (Benchmark)
   - XGB
   - SVM
   - RF

Motivation

- Create a corporate bankruptcy prediction
- Use high dimensional financial data
- Solve class imbalance by creating synthetic samples
- Use feature selection (LASSO, XGBoost)
- Comparing LR, XGB, Random Forest and SVM

Study Plan


Raw Data
• Delisting information
• Financial statement
Data Preprocessing
• Missing value indicator & inputation
• One-hot encoding
• Log transformation
• Scaler
X: 278 variables
Y: delist dummy
X train, Y train
2019-2023
X test, Y test
2024
SMOTE
S(X train, Y train)
Feature Selection
No selection
LASSO
XGBoost
Machine Learning
Logistic regression
Random forest
XGBoost
Support Vector Machine
• NoSe(S(X train, Y train))
• LASSO(S(X train, Y train))
• XGBoost(S(X train, Y train))
Model Evaluation
F1-score
PR-AUC

Data

- 11475 companies
- 118 features
- Large class imbalance

Variable type	Number of type
Categorical	8
Text	6
Continuous	104

Data Information	
Period	2019/1/1-2024/12/31
Frequency	Yearly

Database	
Bankruptcy	WRDS CRSP US Stock delisting information
Financial Statement	WRDS Compustat Fundamental Annual

Delisting Reason Type belongs to BKPY(Bankruptcy)

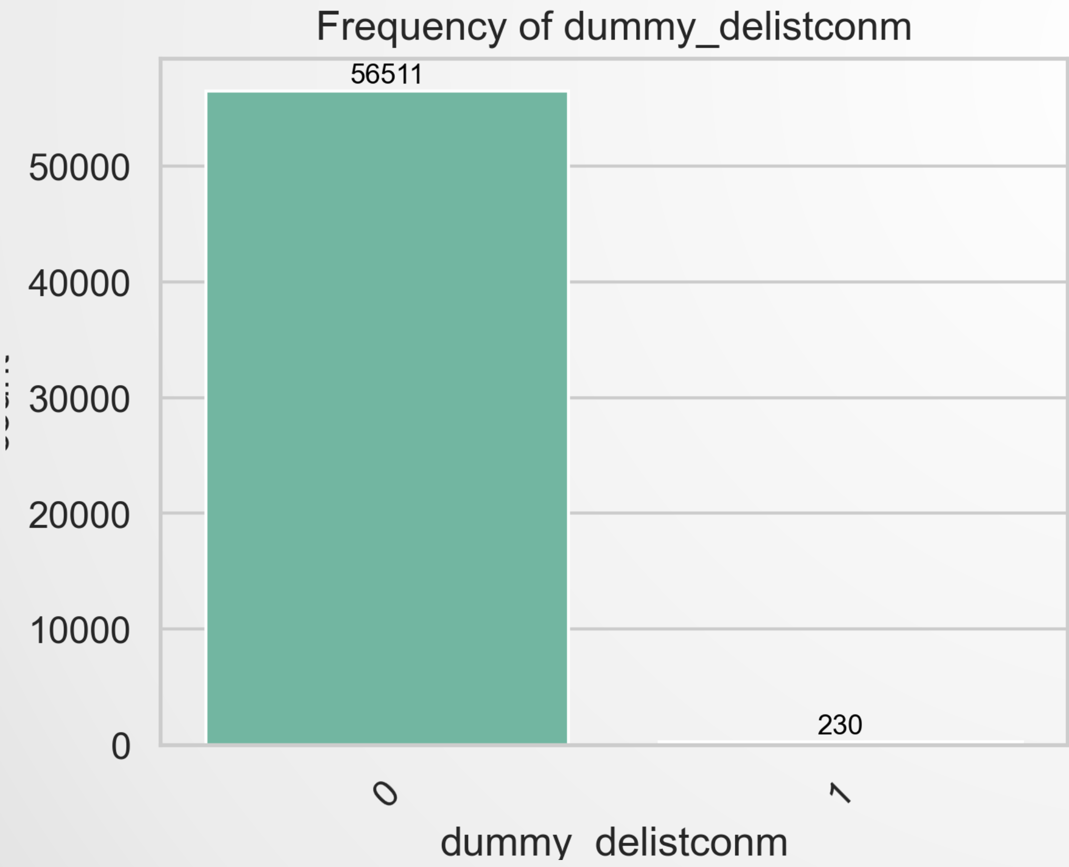
wrds WHARTON RESEARCH DATA SERVICES
The Global Standard for Business Research

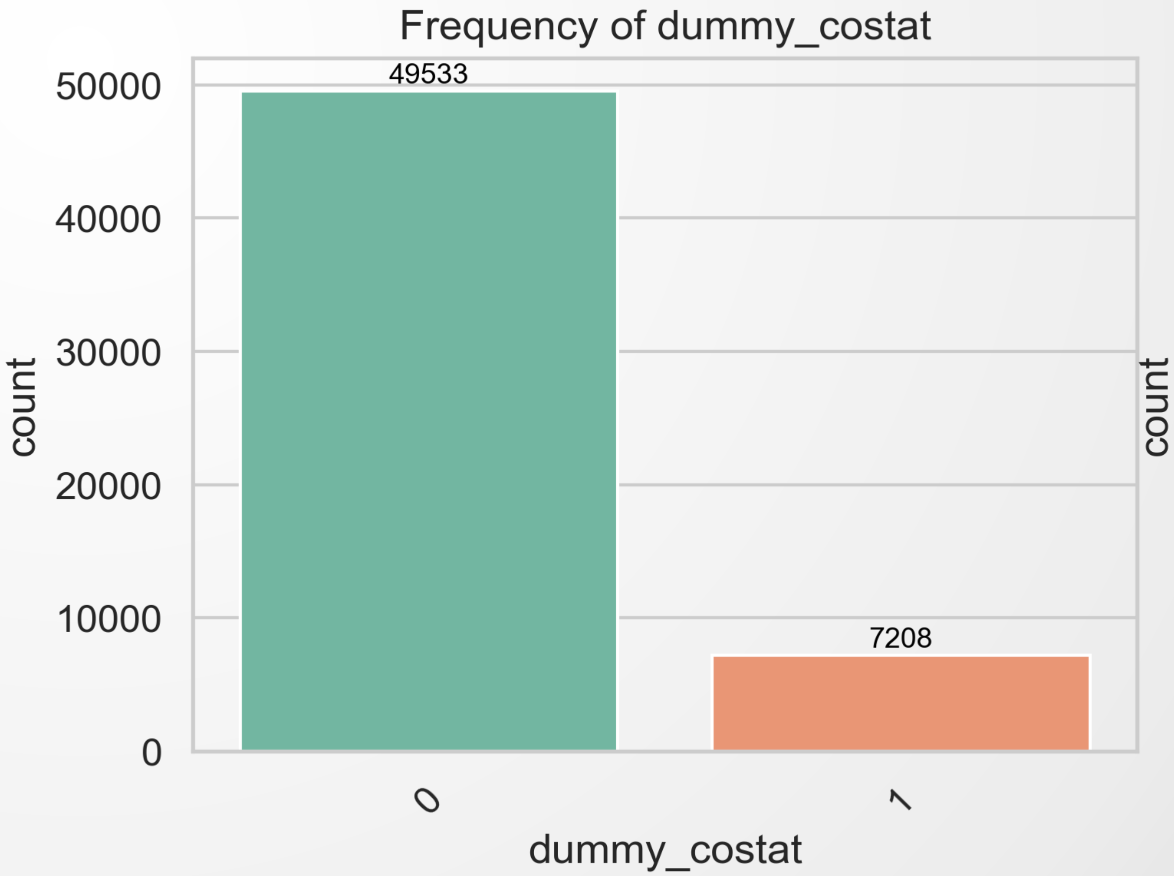
WRDS provides access to standardized, high-quality financial and market datasets that ensure consistency and reliability for empirical research

Data

- 11475 companies
- 118 variables
- Large class imbalance

Dummy Variable	Non-Delist Frequency	Non-Delist (%)	Delist Frequency	Delist (%)
dummy delist company	**56,511**	**99.59%**	**230**	**0.41%**
dummy company status	59,533	87.30%	7,208	12.70%



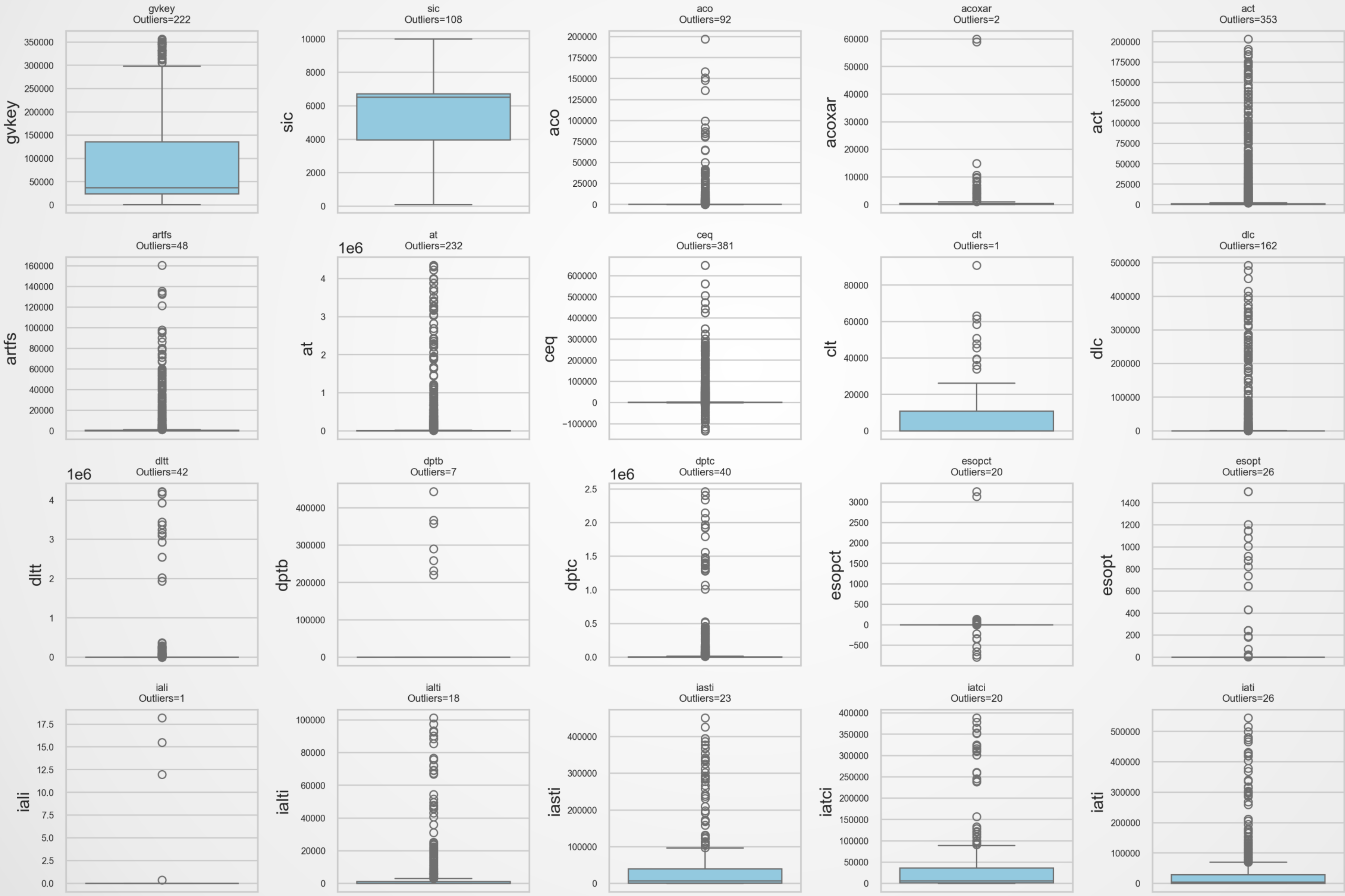


Data pre-processing

Processing types	Effects
Missing value	Fill with use the median for each variable with missing
	Create a dummy missing indicator for each missing value
One-hot encoding	Create/convert a categorical variable into binary (0/1) feature
Log transforming	To deal with right-skewed distribution
Standard scaler	Uniform Feature Scale

Data

- Boxplots of selected features



Data

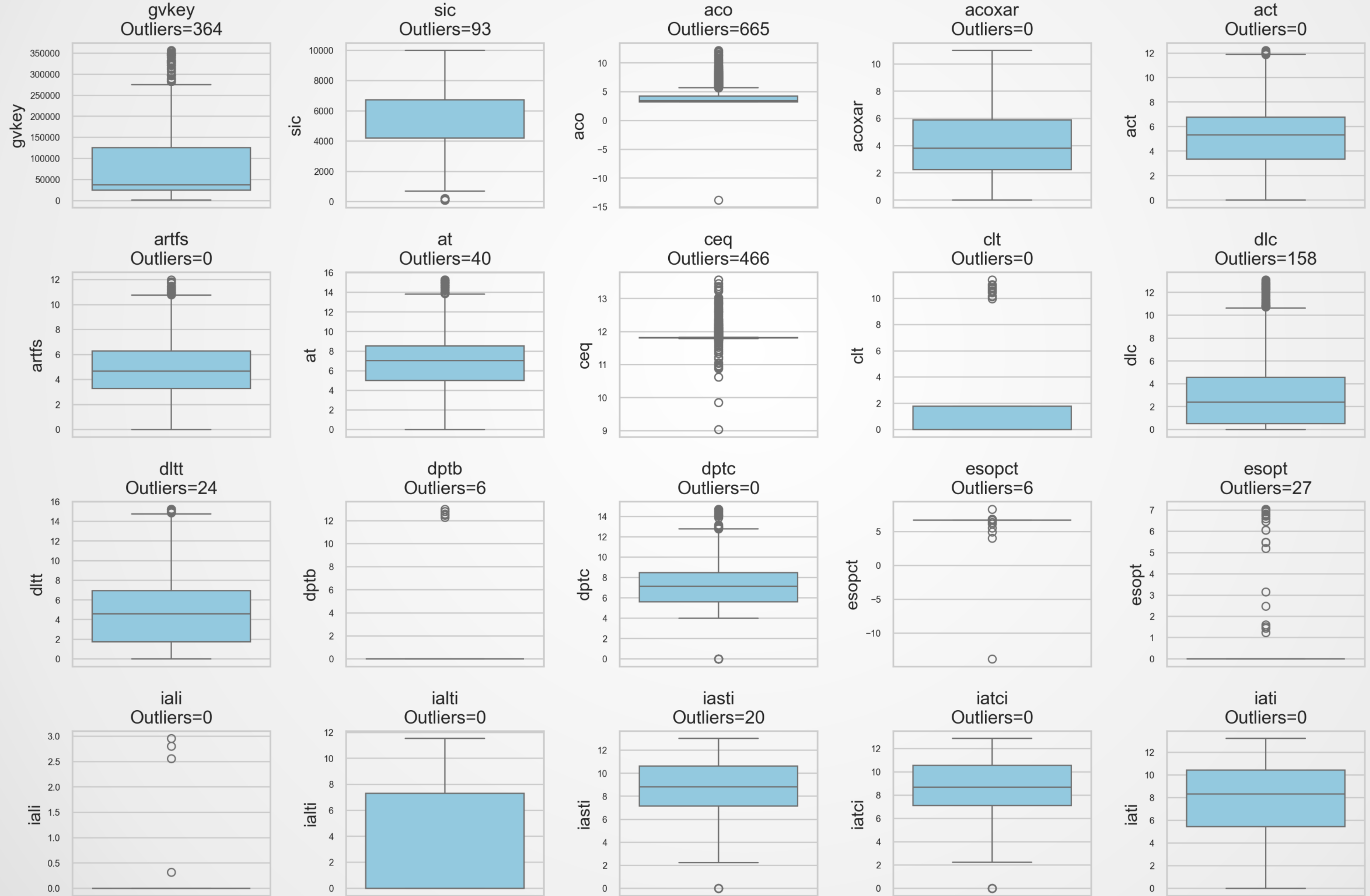
- Description summary for selected numerical features
- Log transformation

	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
lcat	280.00	7286.79	38020.83	0.00	0.00	0.00	0.00	311448.00	6.29	40.18
dvt	42636.00	156.10	835.28	-709.00	0.00	0.00	36.00	25999.00	12.40	203.35
uopi	1414.00	815.65	1124.10	-3221.00	177.00	395.90	1011.00	9231.00	2.67	9.45
nits	462.00	655.20	1288.71	-3090.00	23.50	144.15	848.05	7713.00	2.36	7.61
artfs	2499.00	2334.35	10414.77	0.00	23.93	104.14	514.04	160673.00	8.11	80.88
ialti	575.00	4833.88	14706.52	0.00	0.00	0.00	1225.19	101294.00	4.38	20.20
lcabg	386.00	5304.75	32527.43	0.00	0.00	0.00	0.00	311448.00	7.44	56.90
cnltbl	527.00	2877.22	8324.58	-0.01	0.00	152.32	1296.13	57646.00	4.32	19.85
ppent	41529.00	1740.90	9394.63	0.00	4.06	37.59	344.87	328806.00	14.04	281.17
pltbl	532.00	10105.54	34816.41	-3354.00	0.00	0.00	1942.50	308810.00	5.16	30.80
idit	25643.00	291.47	4182.15	-16.00	0.00	0.60	15.00	193933.00	26.56	829.10
tstk	42740.00	824.64	6587.93	-26.22	0.00	0.00	7.44	254917.00	17.37	424.41
ptbl	568.00	13957.84	34604.84	-3354.00	453.14	1403.50	8188.25	308810.00	4.71	27.43

The entire description summary refers to overleaf table1

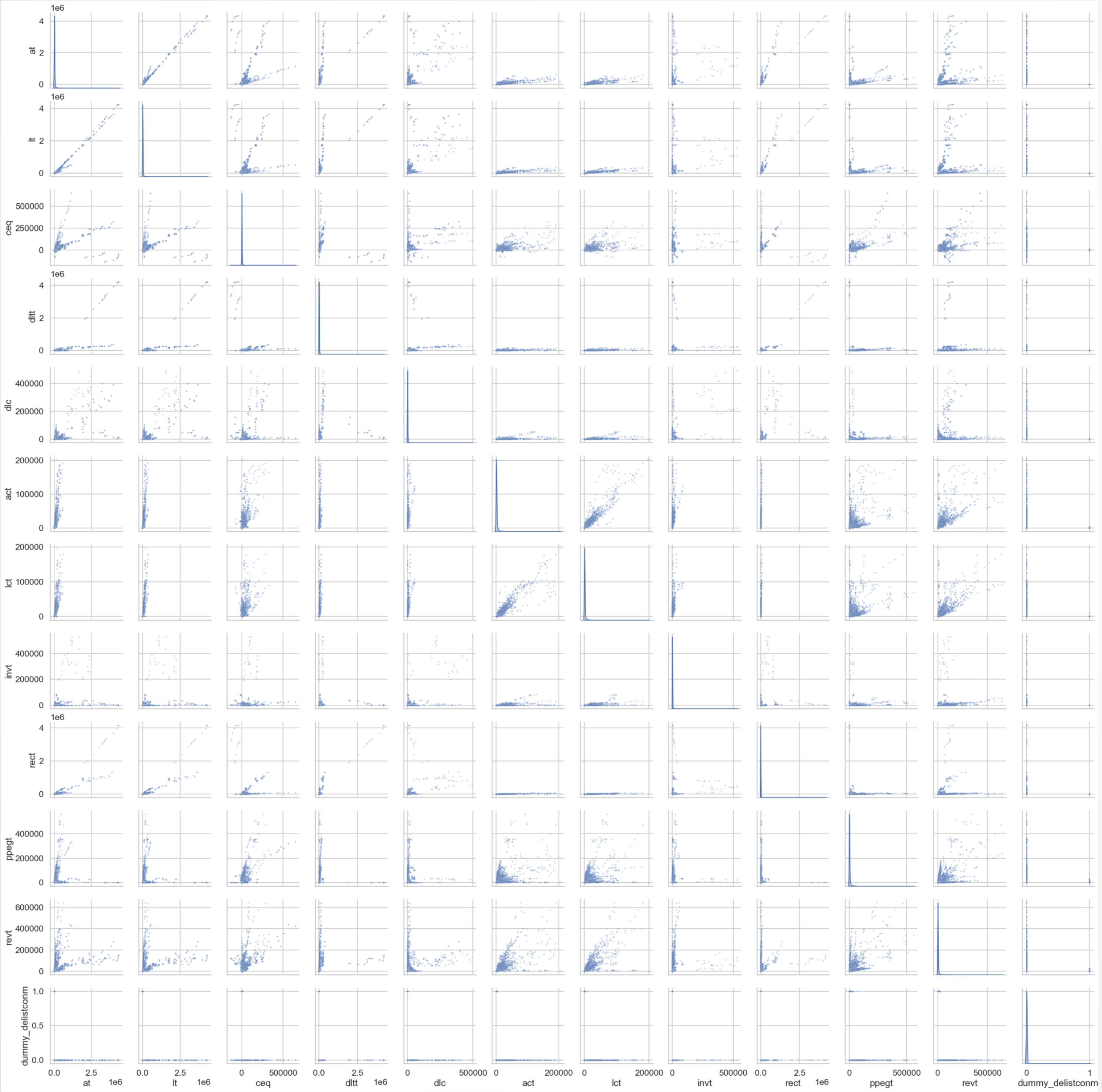
Data

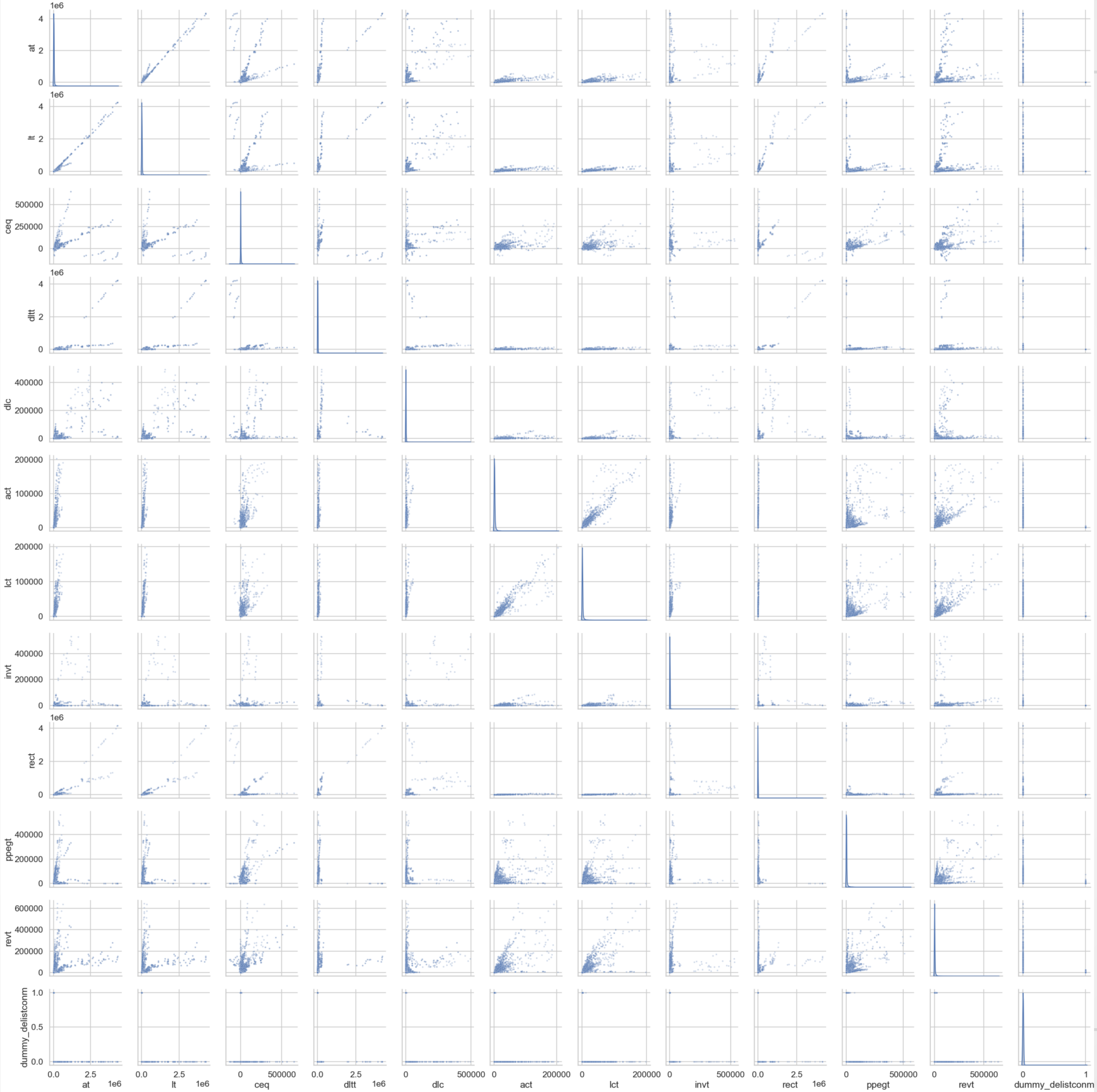
- Boxplots of selected features (after log transform)



Data

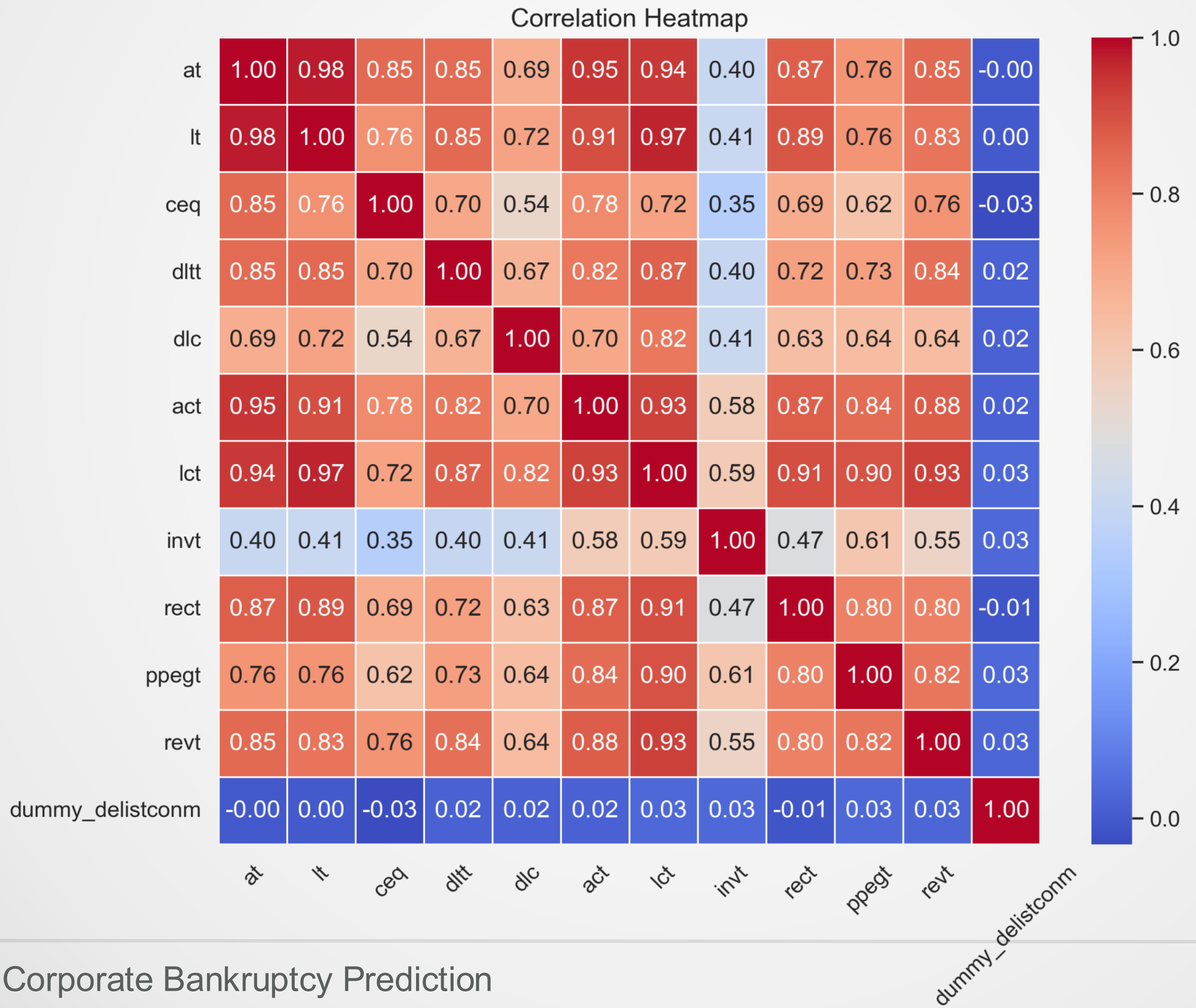
- Pairplot of selected features




1e6
at
lt
ceq
dltt
dlc
act
lct
invt
rect
ppegt
revt
dummy_delistcomm
0
2
4
0.0
2.5
500000
250000
400000
200000
100000
600000
0.5
1.0
1

Data

- Heatmap of selected features



Benchmark

Logistic Regression

- Couronné, Probst & Boulesteix (2018): Suppose we have observations N, for each observation, let: $\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ be the vector of p predictors, and $Y_i \in \{0,1\}$ be the binary outcome. The logistic regression model is defined as:

$$P(Y_i = 1 \mid \mathbf{x}_i) = \pi_i = \frac{1}{1 + \exp\left(-(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})\right)}$$

where:

- π_i is the predicted probability of class 1
- $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$ is the vector of predictor variables
- $(\beta_0, \beta_1, \ldots, \beta_p)$ are the model coefficients to be estimated

Machine Learning Model

Random forest

- Breiman (2001): At a given node t, the Gini impurity is defined as:

$$\text{Gini}(t) = 1 - \sum_{k=1}^{K} p_k(t)^2$$

where:

  - $p_k(t)$ denote the proportion of samples belonging to class k within node t

- Let $T_b(x_i) \in \{0,1\}$ denote the prediction of the b-th tree for sample i. The prediction is obtained through a majority-voting mechanism across the B trees:

$$\hat{y}_i = \text{mode}\{T_1(x_i), T_2(x_i), \ldots, T_B(x_i)\} .$$

Machine Learning Model

XGBoost

- Chen & Guestrin (2016): The model prediction for sample i is obtained by summing the outputs of all regression trees in the ensemble:

$$\hat{y}_i = \sum_{m=1}^{M} f_m(x_i)$$

where:

  - $f_m(\cdot)$ denotes the output of the m-th tree
  - M is the total number of trees
  - x_i is the feature vector of the i-th observation

- Apply the sigmoid transformation to convert the score:

$$p_i = \sigma(\hat{y}_i) = \frac{1}{1 + e^{-\hat{y}_i}}$$

where:

  - p_i represents the predicted probability that observation i belongs to the positive class

Machine Learning Model

SVM

- The model prediction for sample i is obtained by the decision function: $\hat{y}_i = f(x_i) = \mathbf{w}^\top x_i + b$

  for kernel SVM:
  $$\hat{y}_i = \sum_{j=1}^{N} \alpha_j y_j K(x_j, x_i) + b$$

  - x_i feature vector of the i-th observation
  - $\mathbf{w}$ denotes weight vector
  - b is bias term
  - α_j is lagrange multiplier for support vector j
  - $y_j \in \{-1, +1\}$ is class label
  - $K(x_j, x_i)$ is kernel function
  - N denotes number of support vectors

Apply the sigmoid transformation to convert the score into probability

$$p_i = \sigma(\hat{y}_i) = \frac{1}{1 + e^{-A\hat{y}_i - B}}$$

Feature Selection Model

LASSO

- Logistic LASSO: applying an L1 penalty that forces uninformative or redundant coefficients to zero

$$\hat{\beta} = \arg\min_{\beta} \left\{ -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right] + \lambda \sum_{j=1}^{p} |\beta_j| \right\}, \qquad p_i = \frac{1}{1+\exp(-\mathbf{x}_i^{\top}\beta)}.$$

Where:

- $p_i = \frac{1}{1+\exp(-x_i^{\top}\beta)}$ denotes the predicted probability
- $||\beta||_1 = \sum_{j=1}^{p} |\beta_j|$ is the L_1 norm, and $\lambda > 0$ controls the regularization strength

Feature Selection Model

XGBoost

- XGBoost: the average improvement in the model's objective when the feature is used for splitting.

$$\text{Gain} = \frac{1}{2}\left(\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{G^2}{H+\lambda}\right) - \gamma \qquad \text{Importance}(f) = \frac{1}{S}\sum_{s=1}^{S} \text{Gain}_s(f)\,.$$

where:

- λ is the L_2 regularization parameter
- γ is the penalty for creating a new leaf
- f is computed as the average gain across all splits where f is used

Results

Classifier	Accuracy	AUC	Recall (1)	Precision (1)	F1 (1)	AP
XGB	0.9980	0.9769	0.4167	0.2778	0.3333	0.371
SVM	0.9945	0.9948	**0.8333**	0.1613	**0.2703**	**0.373**
RF	0.9853	0.9469	0.1667	0.0147	0.0270	0.113
LR	0.9845	0.9551	0.5833	0.0455	0.0843	0.114

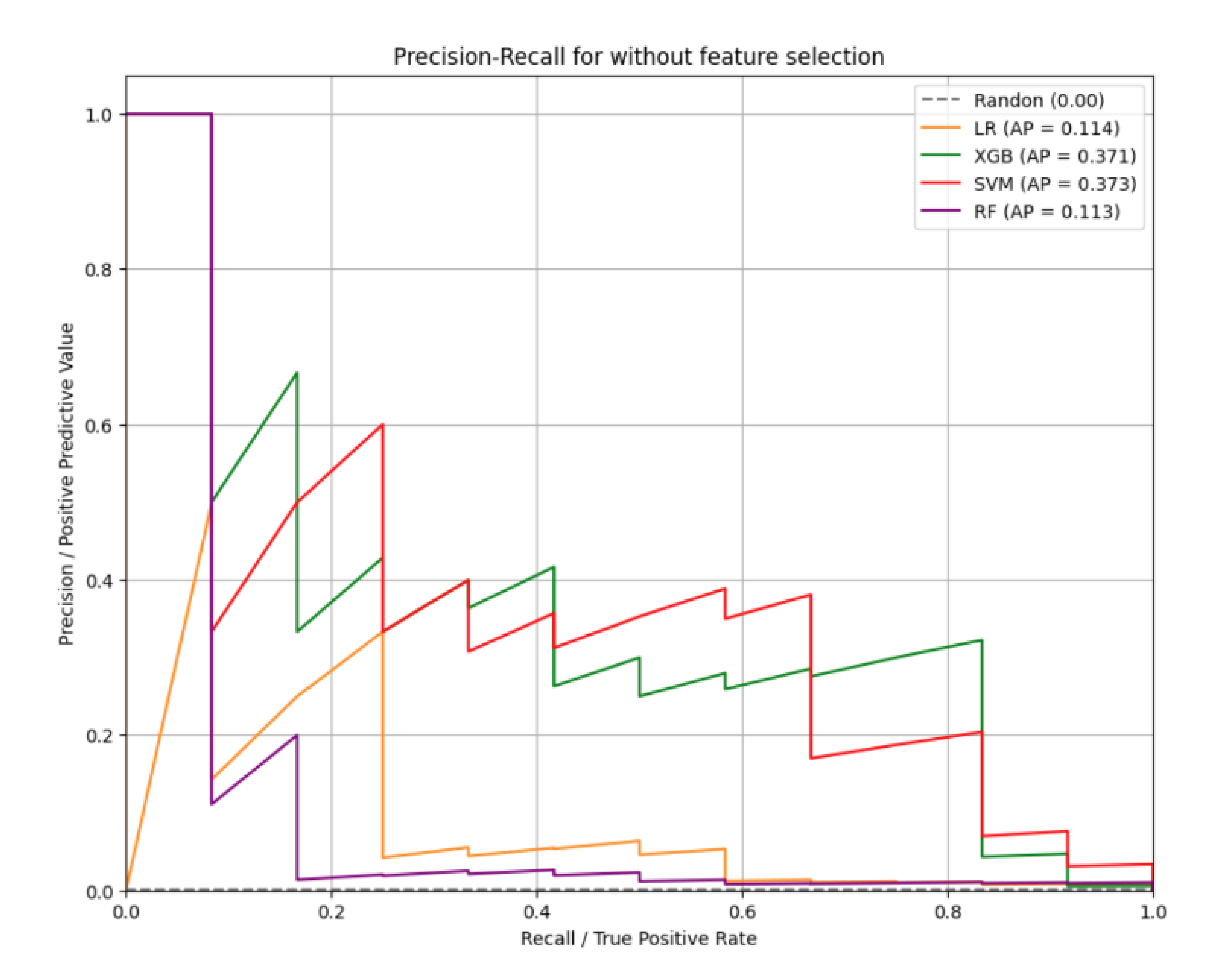
Panel A: LASSO feature selection

Classifier	Accuracy	AUC	Recall (1)	Precision (1)	F1 (1)	AP
XGB	0.9982	0.9877	0.6667	0.3636	0.4706	**0.420**
SVM	0.9954	0.9942	**0.7500**	0.1765	0.2857	0.335
RF	0.9988	0.9834	0.3333	**0.5000**	0.4000	0.305
LR	0.9829	0.9630	0.5833	0.0400	0.0773	0.210

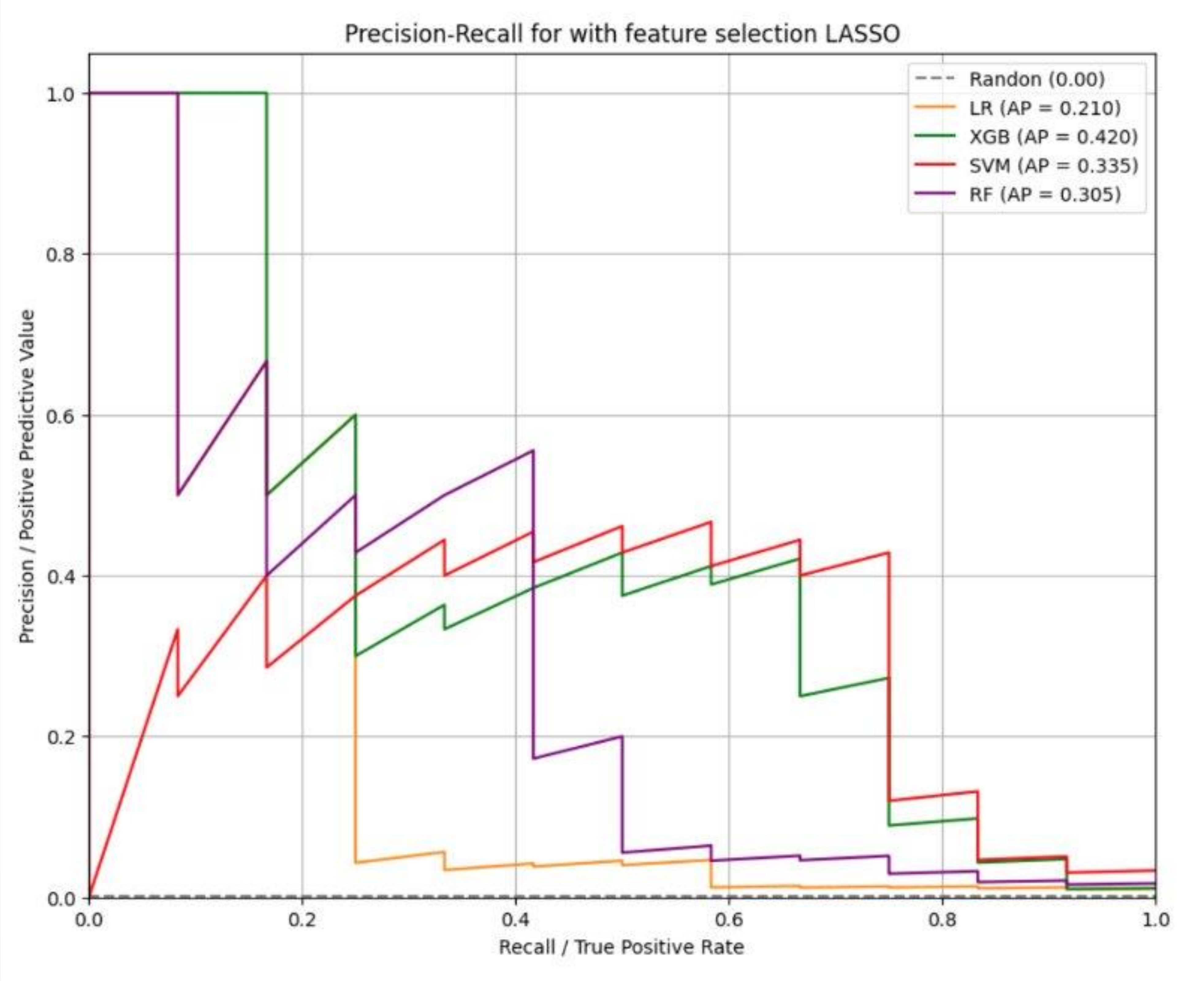
Panel B: XGBoost feature selection

Classifier	Accuracy	AUC	Recall (1)	Precision (1)	F1 (1)	AP
XGB	0.9989	0.9800	0.500	0.5500	0.5200	**0.541**
SVM	0.9919	0.9820	0.6700	0.1000	0.1700	0.224
RF	0.9982	0.9670	0.2500	0.2500	0.250	0.242
LR	0.9732	0.9430	0.5833	0.0265	0.0500	0.064

Results

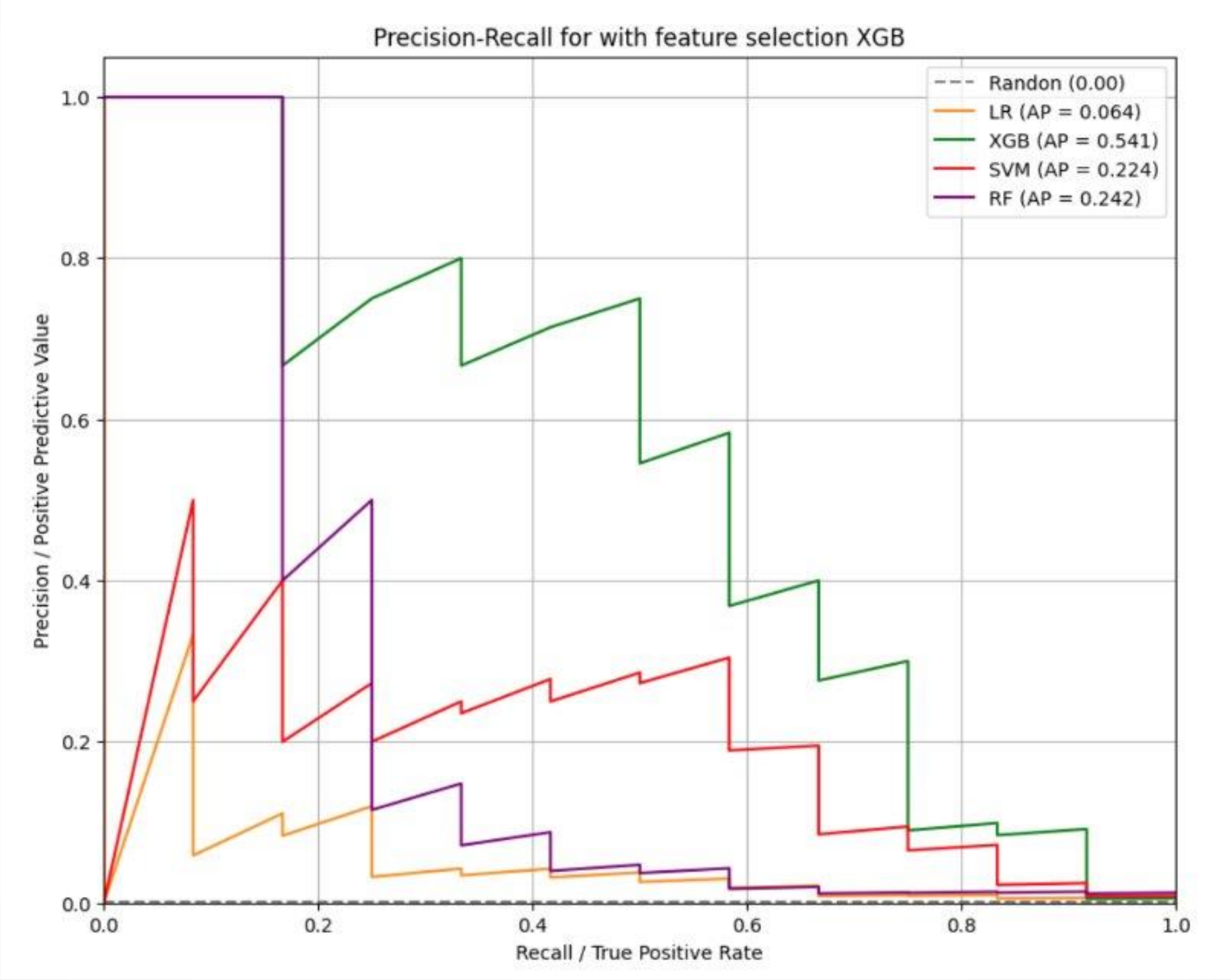

Precision-Recall for without feature selection
Randon (0.00)
LR (AP = 0.114)
XGB (AP = 0.371)
SVM (AP = 0.373)
RF (AP = 0.113)
Precision / Positive Predictive Value
Recall / True Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

Results



Precision-Recall for with feature selection LASSO
Randon (0.00)
LR (AP = 0.210)
XGB (AP = 0.420)
SVM (AP = 0.335)
RF (AP = 0.305)
Precision / Positive Predictive Value
Recall / True Positive Rate
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

Results


Precision-Recall for with feature selection XGB
Randon (0.00)
LR (AP = 0.064)
XGB (AP = 0.541)
SVM (AP = 0.224)
RF (AP = 0.242)
Precision / Positive Predictive Value
Recall / True Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

Conclusion

- XGBoost performs more stable under feature selection and no selection
- SVM has higher recall but lower precision, and with little improvement after feature selection
- RF and XGB perform better with feature selection
- Missing-value indicators disclosure as a distress signal
- Feature selection is essential for extracting predictive signals from high-dimensional financial data

Model input and output

LASSO feature selection

Symbol	Acronym / Variable name	Description
Y	dummy_delistconm	Equals 1 if the firm experienced a delisting event; 0 otherwise.
X_1	at	Total assets (firm size proxy).
X_2	lt	Total liabilities (financial obligations).
X_3	ceq	Common equity (book value of shareholders equity).
X_4	dltt	Long-term debt (noncurrent portion of total debt).
X_5	dlc	Debt in current liabilities (short-term borrowings).
X_6	act	Current assets (short-term resources).
X_7	lct	Current liabilities (short-term obligations).
X_8	invt	Inventories (goods held for sale or production).
X_9	rect	Accounts receivable (customer credit sales).
X_{10}	ppegt	Property, plant, and equipment (gross).
X_{11}	revt	Total revenue (gross sales).
X_{12}	nit	Net income (profit after taxes).
X_{13}	xopr	Operating expenses (total).
X_{14}	xint	Interest and related expenses (cost of debt).
X_{15}	txt	Income taxes (total).
X_{16}	intan	Intangible assets (goodwill and similar items).
X_{17}	ivst	Short-term investments (marketable securities).
X_{18}	fopt	Funds from operations (cash generated internally).
X_{19}	fsrct	Sources of funds (cash inflows).
X_{20}	fuset	Uses of funds (cash outflows).
X_{21}	dummy_costat	Equals 1 if the firm is classified as inactive in Compustat; 0 otherwise.
$X_{22}X_n$	Industry dummies	Binary indicators for industry sectors (SIC-based).
$X_{n+1}X_p$	Missing indicators	Dummy variables indicating missing accounting items.

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